

Tutorial 9

Basis, dimension and subspaces associated with a matrix

1. Suppose that $V = \mathbb{R}^m$. For each of the following statements say whether it is true or false, and give a reason for your answer. You may quote without proof any theorem from the notes. (This should also go quickly!)

- (a) Every basis for V has m vectors.

Solution: True. See Theorem 2.5.15 and Example 2.5.17(b).

- (b) Every set of k vectors of V with $k < m$ is linearly independent.

Solution: False. For example, in $\mathbb{R}^3$, the set $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \right\}$ is linearly dependent.

- (c) Every set of k vectors of V with $k > m$ is linearly dependent.

Solution: True. See Theorem 2.5.13(a).

- (d) Every set of k vectors of V with $k < m$ fails to span V .

Solution: True. See Theorem 2.5.13(b).

- (e) Every set of k vectors of V with $k > m$ spans V .

Solution: False. For example, in $\mathbb{R}^2$, the set $\left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 9 \end{pmatrix} \right\}$ spans a straight line through the origin, not all of $\mathbb{R}^2$.

2. (a) Let $C = \{x^3, x^2 + 1, x\}$.
i. Is C a basis for P_3 ? Explain briefly why or why not.

Solution: C is not a basis for P_3 since P_3 has dimension 4, and C has only 3 vectors, so C does not span P_3 .

- ii. If C is not a basis for P_3 , can it be enlarged to a basis by adding appropriate vectors? If it can be enlarged to a basis, do this; otherwise explain why it cannot be done.

Solution: $\{x^3, x^2 + 1, x, 1\}$ is a suitable enlargement.

(b) Let $C = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$

- i. Is C a basis for $\mathbb{R}^4$? Explain briefly why or why not.

Solution: C is not a basis for $\mathbb{R}^4$. C is linearly dependent: its third vector is the sum of the first two. Also C has only 3 vectors, whereas $\mathbb{R}^4$ has dimension 4, so C does not span $\mathbb{R}^4$.

- ii. If C is not a basis for $\mathbb{R}^4$, can it be enlarged to a basis by adding appropriate vectors? If it can be enlarged to a basis, do this; otherwise explain why it cannot be.

Solution: C cannot be enlarged to a basis for $\mathbb{R}^4$: any set which contains C will be linearly dependent.

(c) Let $C = \left\{ \begin{pmatrix} 0 & 0 \\ 0 & 7 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 7 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 15 & 6 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \right\}.$

- i. Is C a basis for $M_{2 \times 2}(\mathbb{R})$? Explain briefly why or why not.

Solution: C is not a basis for $M_{2 \times 2}(\mathbb{R})$ since $M_{2 \times 2}(\mathbb{R})$ has dimension 4 and there are 6 vectors in C , so C is linearly dependent.

- ii. If C is not a basis for $M_{2 \times 2}(\mathbb{R})$, can it be reduced to a basis by omitting appropriate vectors? If it can be reduced to a basis, do this; otherwise explain why it cannot be.

Solution: $\left\{ \begin{pmatrix} 0 & 0 \\ 0 & 7 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \right\}$ is a suitable reduction.

3. The trace of an $n \times n$ matrix $A = (a_{ij})$, denoted by $\text{tr}(A)$, is the sum of the entries on the diagonal of A , i.e. .

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

- (a) Show that $S = \{A \in M_{n \times n}(\mathbb{R}) : \text{tr}(A) = 0\}$ is a subspace of $M_{n \times n}(\mathbb{R})$

Solution: S is nonempty, closed under addition, and closed under scalar multiplication. Hence S is a subspace of $M_{n \times n}(\mathbb{R})$.

- (b) Find a basis for S .

Solution: Let E_{ij} be the matrix with 1 in the (i, j) entry and 0 everywhere else. Then a basis for S is for instance the set

$$\{E_{ij} \mid i \neq j\} \cup \{E_{ii} - E_{nn} \mid i < n\}.$$

- (c) What does $\dim(S)$ equal?

Solution: The dimension of S is $n^2 - 1$.

4. Let $A = \begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 3 & 7 & 10 & 5 & 5 \\ 1 & 3 & 4 & 0 & 3 \\ 6 & 15 & 21 & 7 & 12 \end{pmatrix}.$

We can row reduce A to U , where $U = \begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 1 & -1 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$

(You do not need to check this!) Which of the following statements are true? (You should be able to answer this question very quickly; **use your notes.**)

- (a) $NS(A) = NS(U)$.
- (b) $RS(A) = RS(U)$.
- (c) $CS(A) = CS(U)$.
- (d) $\text{rank}(A) = \text{rank}(U)$
- (e) $\text{nullity}(A) = \text{nullity}(A^T)$

Solution: (a), (b) and (d) are true.

5. Let A be the matrix in Question 4. Answer the following questions, using where appropriate, the fact that A can be row-reduced to the matrix U in Question 4.

- (a) Write down a basis for, and the dimension of, $NS(A)$.

Solution: Basis for $NS(A) = \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ and $\dim NS(A) = 2$.

(b) Write down a basis for, and the dimension of, $RS(A)$.

Solution: Basis for $RS(A) = \{(1, 2, 3, 2, 1), (0, 1, 1, -1, 2), (0, 0, 0, 1, 0)\}$ and $\dim RS(A) = 3$.

(c) Write down a basis for, and the dimension of, $CS(A)$.

Solution: Basis for $CS(A) = \left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \\ 6 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \\ 3 \\ 15 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 0 \\ 7 \end{pmatrix} \right\}$ and $\dim CS(A) = 3$.

(d) What is the nullity of A ?

Solution: Nullity(A) is 2.

(e) What is the rank of A ?

Solution: rank(A)=3.

(f) What is the nullity of A^T ?

Solution: nullity(A^T) = $4 - \text{rank}(A^T) = 4 - \text{rank}(A) = 4 - 3 = 1$.

6. (a) Suppose A is a 11×46 matrix.

- (i) What is the maximum possible value for the rank of A ?
- (ii) What is the minimum possible value for the rank of A ?
- (iii) What is the maximum possible value for the nullity of A ?
- (iv) What is the minimum possible value for the nullity of A ?

Solution: (i) 11 (ii) 0 (iii) 46 (iv) 35

- (b) Answer the same questions as (a) for a 46×11 matrix A .

Solution: (i) 11 (ii) 0 (iii) 11 (iv) 0

7. Let V be a vector space and B be a basis for V .

- (a) Prove that if S is a linearly independent subset of V which contains B , then $S = B$. (This is sometimes expressed by saying a basis is a *maximal linearly independent* subset.)

Solution: We have that $B \subseteq S$. If $B \neq S$, S would be a linearly independent set with more vectors than B , contradicting Theorem 2.5.13(a).

- (b) Prove that if S is a subset of V which spans V and is contained in B , then $S = B$. (This is sometimes expressed by saying a basis is a *minimal spanning* subset.)

Solution: In this case we have $S \subseteq B$. If $S \neq B$, S would be a set with fewer vectors than B which spans V , contradicting Theorem 2.5.13(b).

- (c) Prove that a maximal linearly independent subset of V is a basis for V .

Solution: Let S be a maximal linearly independent set in V . We need to show that $\text{span } S = V$. If not, there will be a $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span } S$. Then it follows from Theorem 2.5.19(a) that $S \cup \{\mathbf{v}\}$ is linearly independent, contradicting the fact that S is a maximal linearly independent set.

- (d) Prove that a minimal spanning subset of V is a basis for V .

Solution: The proof is similar to (c), but uses Theorem 2.5.19(b).